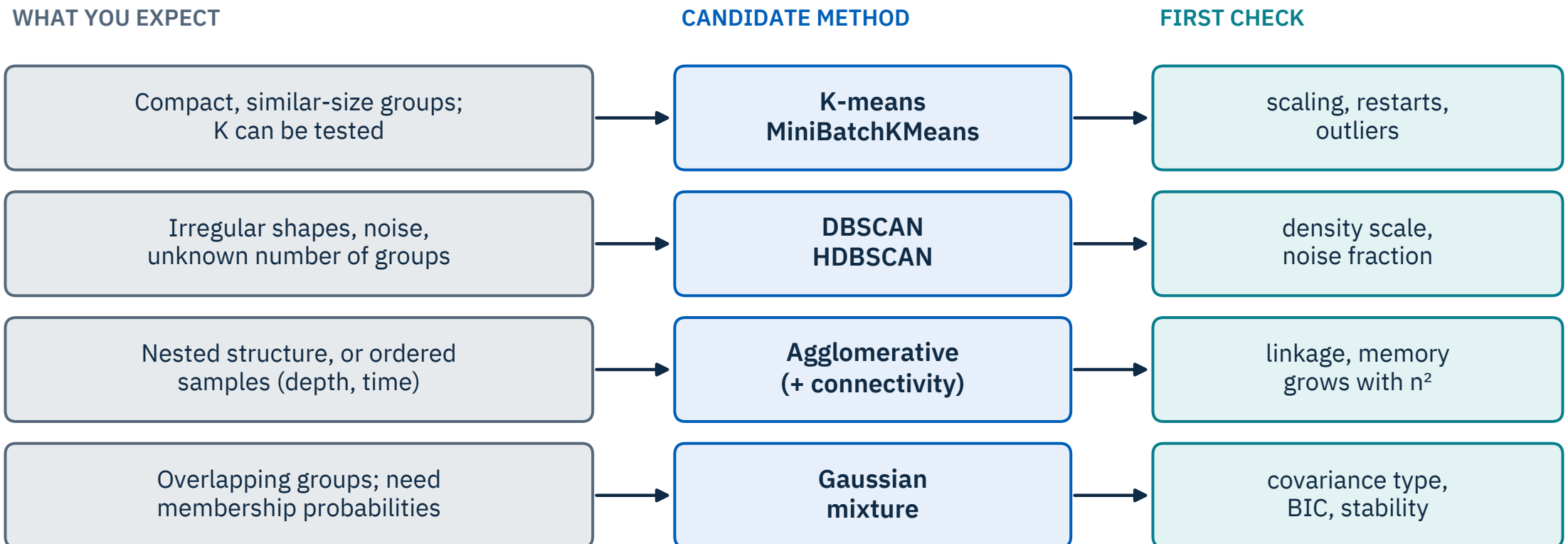
WHAT YOU EXPECT
CANDIDATE METHOD
FIRST CHECK
Compact, similar-size groups; K can be tested
K-means MiniBatchKMeans
scaling, restarts, outliers
Irregular shapes, noise, unknown number of groups
DBSCAN HDBSCAN
density scale, noise fraction
Nested structure, or ordered samples (depth, time)
Agglomerative (+ connectivity)
linkage, memory grows with n²
Overlapping groups; need membership probabilities
Gaussian mixture
covariance type, BIC, stability